RNA-Seq

Power Calculation prior to experiment

Paired-end sequencing always recommended

Low tumor purity requires more reads

Study Type	Sequencing Depth	Biological Replicates	Technical Replicates	Why?
Differential Expression (Moderate to large difference)	15M-25M	Ideally 6+, minimum of 3	Ideally 3, minimum 2	• 15-25M reads captures differences between highly expressed genes. • More replicates = more statistical power. • Technical replicates account for technical variability.
Differential Expression (Small difference)	30M-50M	Ideally 6+	Ideally 3, minimum 2	• When expecting a smaller effect size between conditions, more replicates and more depth provides more statistical power.
Transcript Isoform Discovery/Splicing analysis	100M ideally, 50M minimum (Paired-end)	Ideally 6+,	Ideally 3, minimum 2	• Paired end allows for detection of splicing events. • High sequencing depth and many replicates needed since measuring transcript abundance instead of gene expression
Differential Expression (lncRNA/low expressing genes)	100M ideally, 50M minimum	Ideally 6+	Ideally 3, minimum 2	• High sequencing depth needed since expression is low in targeted genes

ChIP-Seq

Power Calculation prior to experiment

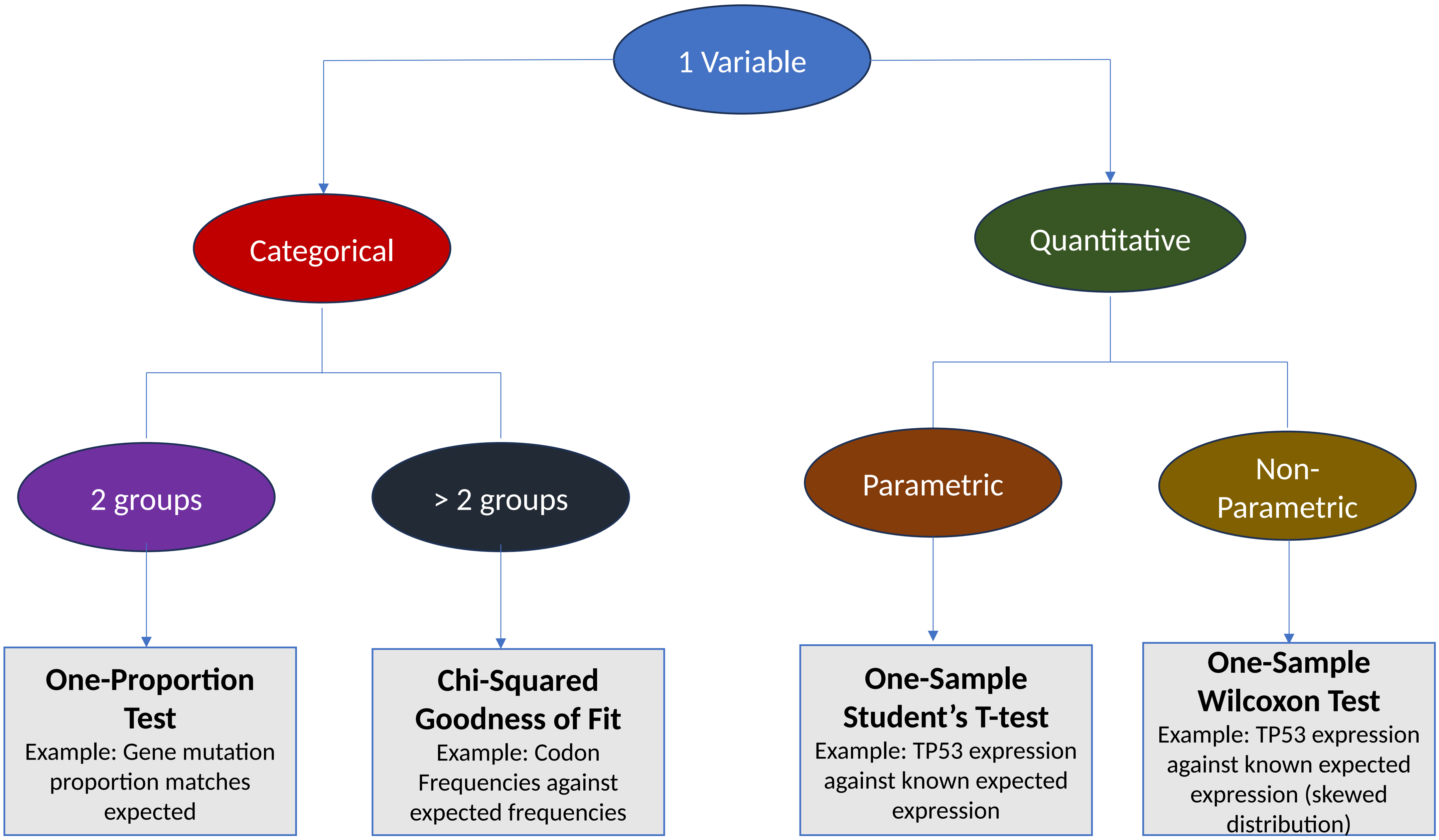
Input depth should be twice as high as ChIP depth

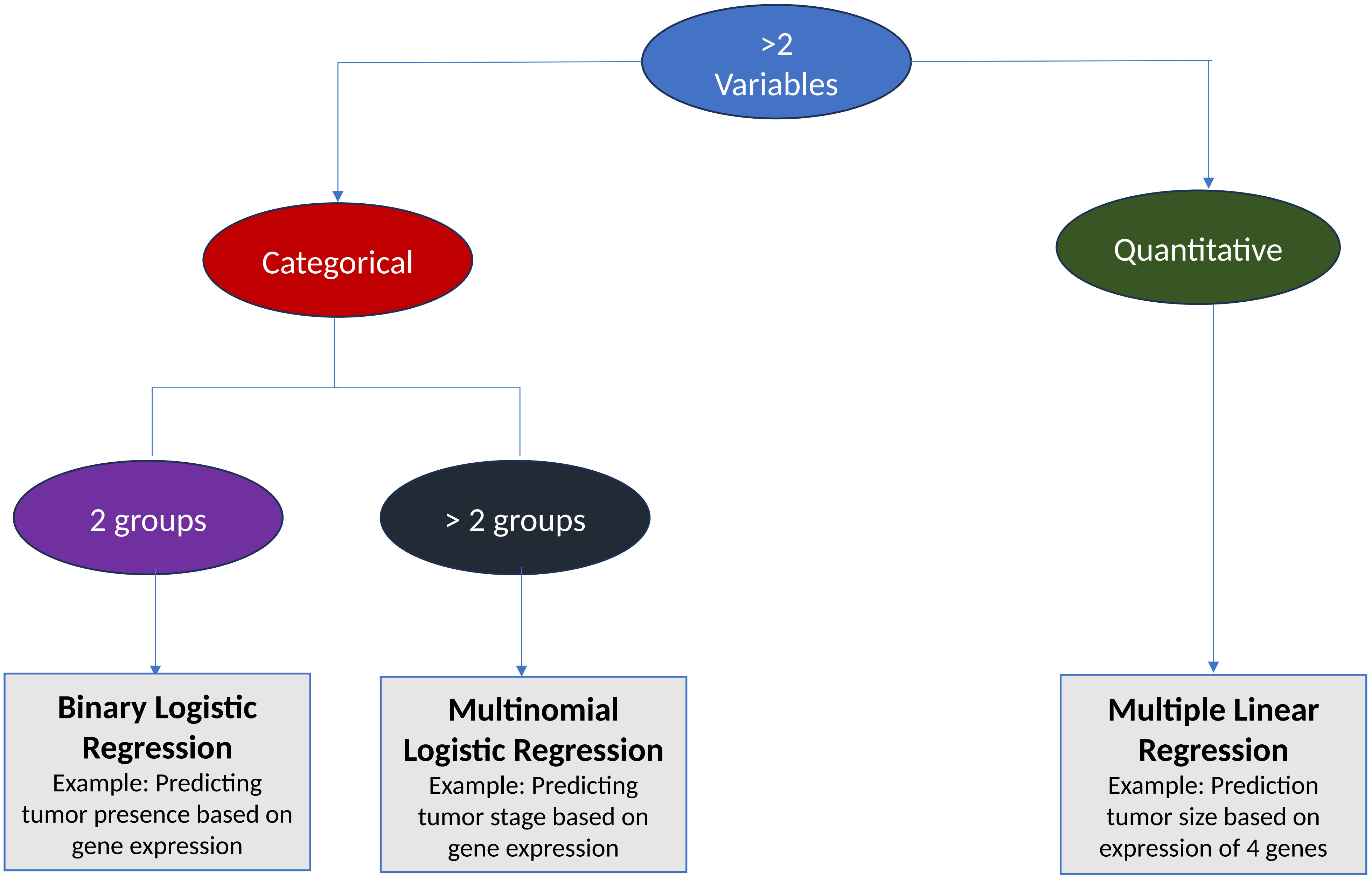
Study Type	Peak Type	Sequencing Depth	Biological Replicates	Technical Replicates	Input Required?	Why?
TF Binding	Narrow	20M-30M	Ideally 3+	Minimum 2	Yes	• TF binding sites typically produce sharp peaks and require lower sequencing depth than other ChIP-seq protocols. • Biological and technical replicates allow for stronger statistical power and control for technical variability
Broad Histone Marks	Broad	50M-100M	Ideally 3+	Minimum 2	Yes	• Broad histone marks cover large regions, requiring higher sequencing depth and more samples for statistical power
Narrow Histone Marks	Narrow	20M-50M (depending on histone mark)	Ideally 3+	Minimum 2	Yes	• Narrow histone marks cover regions of smaller regions than broad sequencing depth, requiring lower sequencing depth and less samples for statistical power than broad histone marks
Subtle/low binding patterns	Broad or narrow	75M+	Ideally 3+	Minimum 2	Yes	• Subtle peaks require a very high sequencing depth since there won’t be as many reads mapping to these peaks.

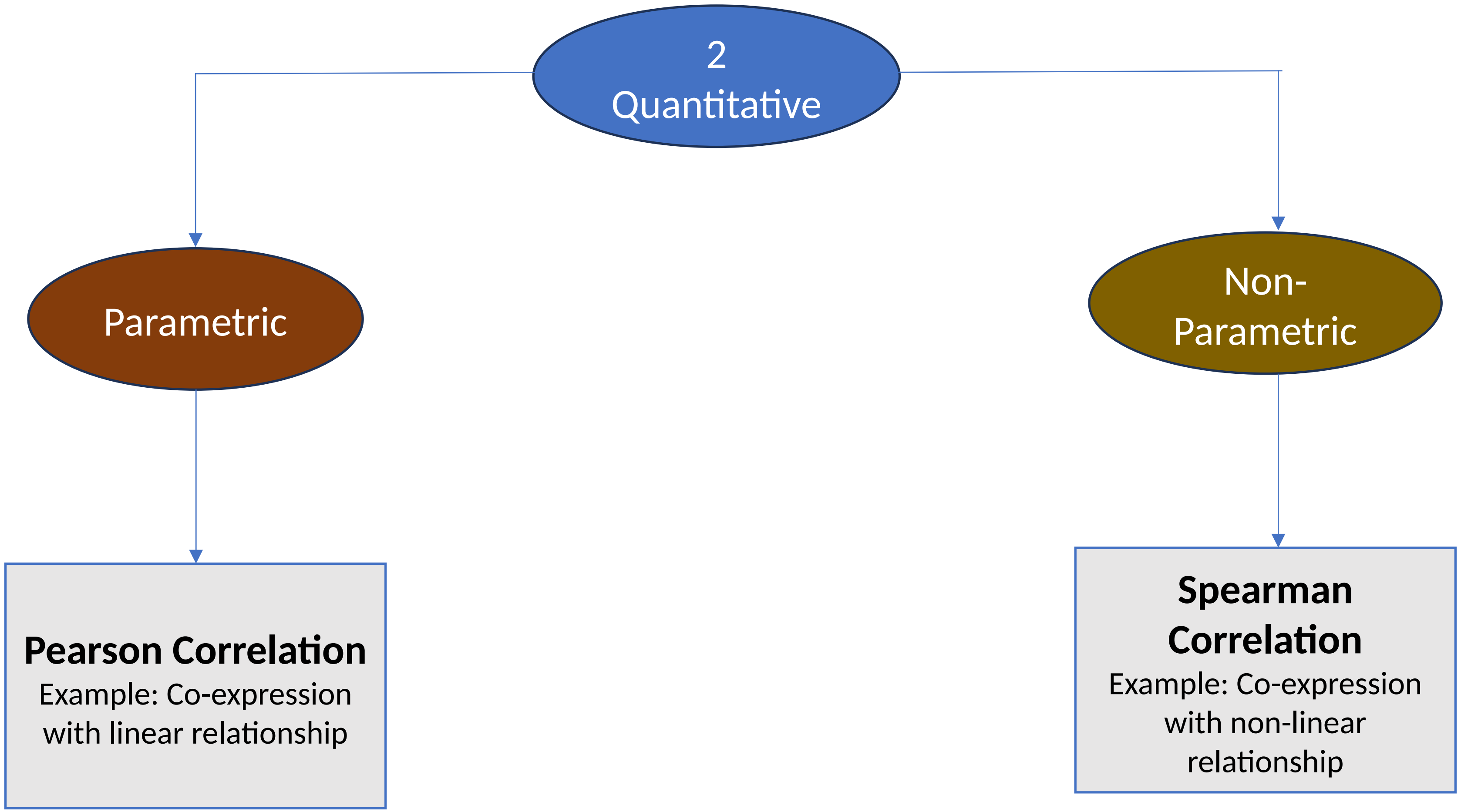
ATAC-Seq

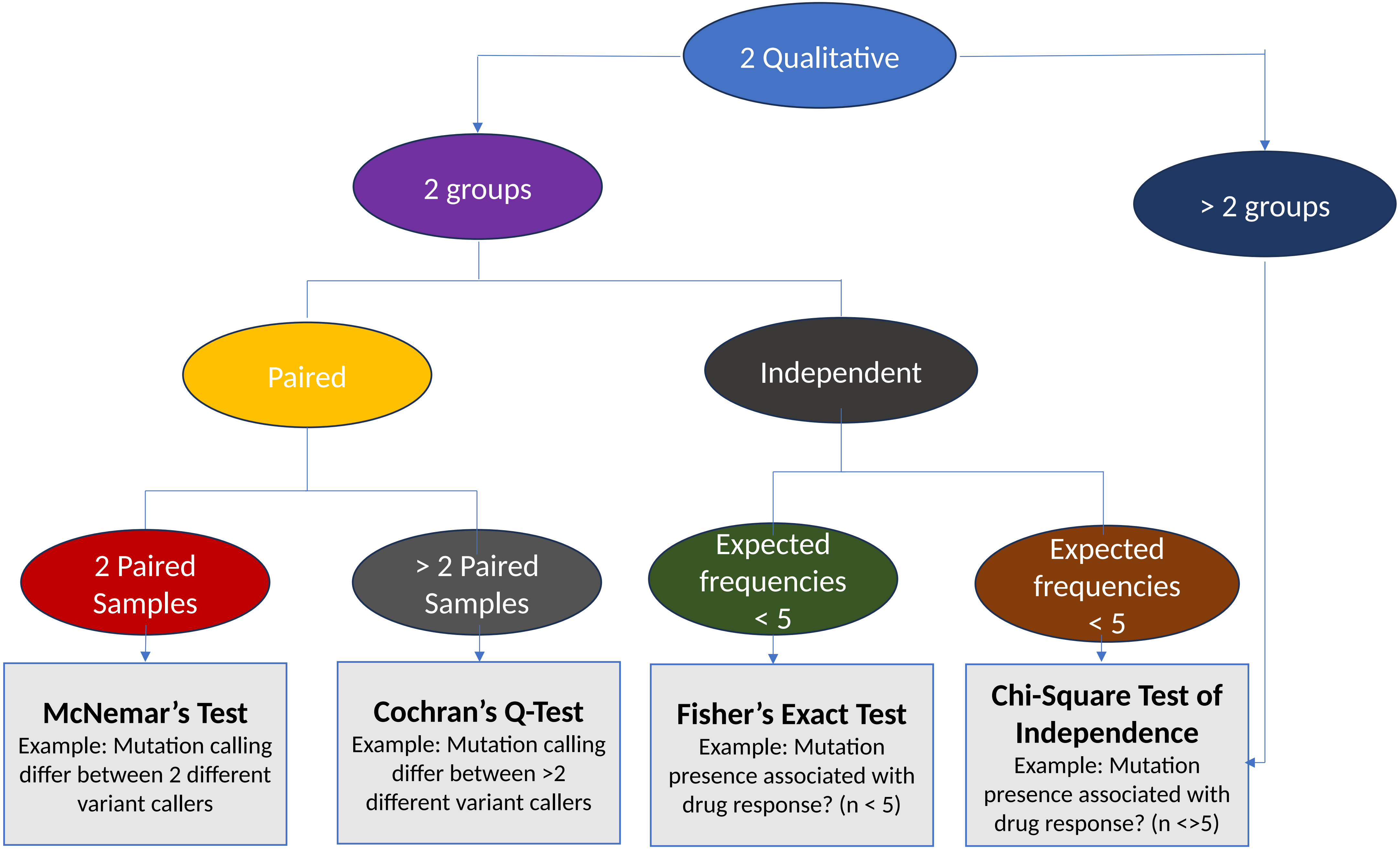
Power Calculation prior to experiment
Fragment length < 100 = Nucleosome free
Fragment length ~ 150-200 = Mono-nucleosome bound
Fragment length >= 300 = Di/Multi-nucleosome bound

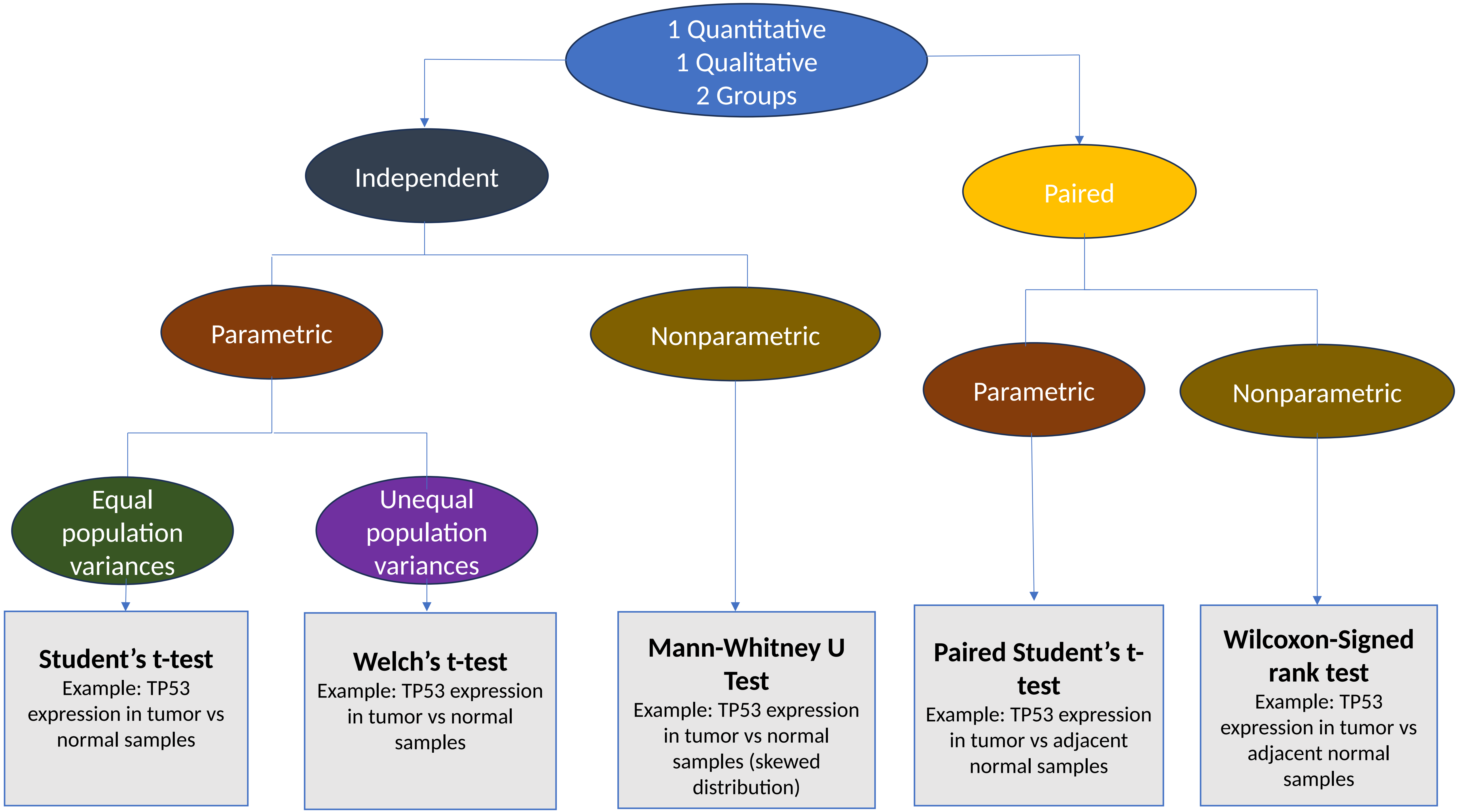
Study Type	Peak Type	Sequencing Depth	Biological Replicates	Technical Replicates	Input Required?	Why?
General Chromatin Accessibility	Broad/Narrow	50M	Ideally 3+	Minimum 2	No	• ATAC-seq requires more reads than ChIP since it captures all open-chromatin regions. • Replicates and higher sequencing depth improves statistical power
Active Regulatory Elements	Narrow	50M-100M	Ideally 3+	Minimum 2	No	• Enhancers and promotors can be dynamic with weak signals, requiring higher sequencing depth and more replicates.
TF Footprinting	Narrow	100M+	Ideally 3+	Minimum 2	No	• TF footprints often have a very small signal size and require single nucleotide resolution, so read depth must be very high.
Differential peaks	Broad or narrow	75M+	Ideally 3+	Minimum 2	No	• Differences between conditions can be very subtle, requiring a higher number of reads

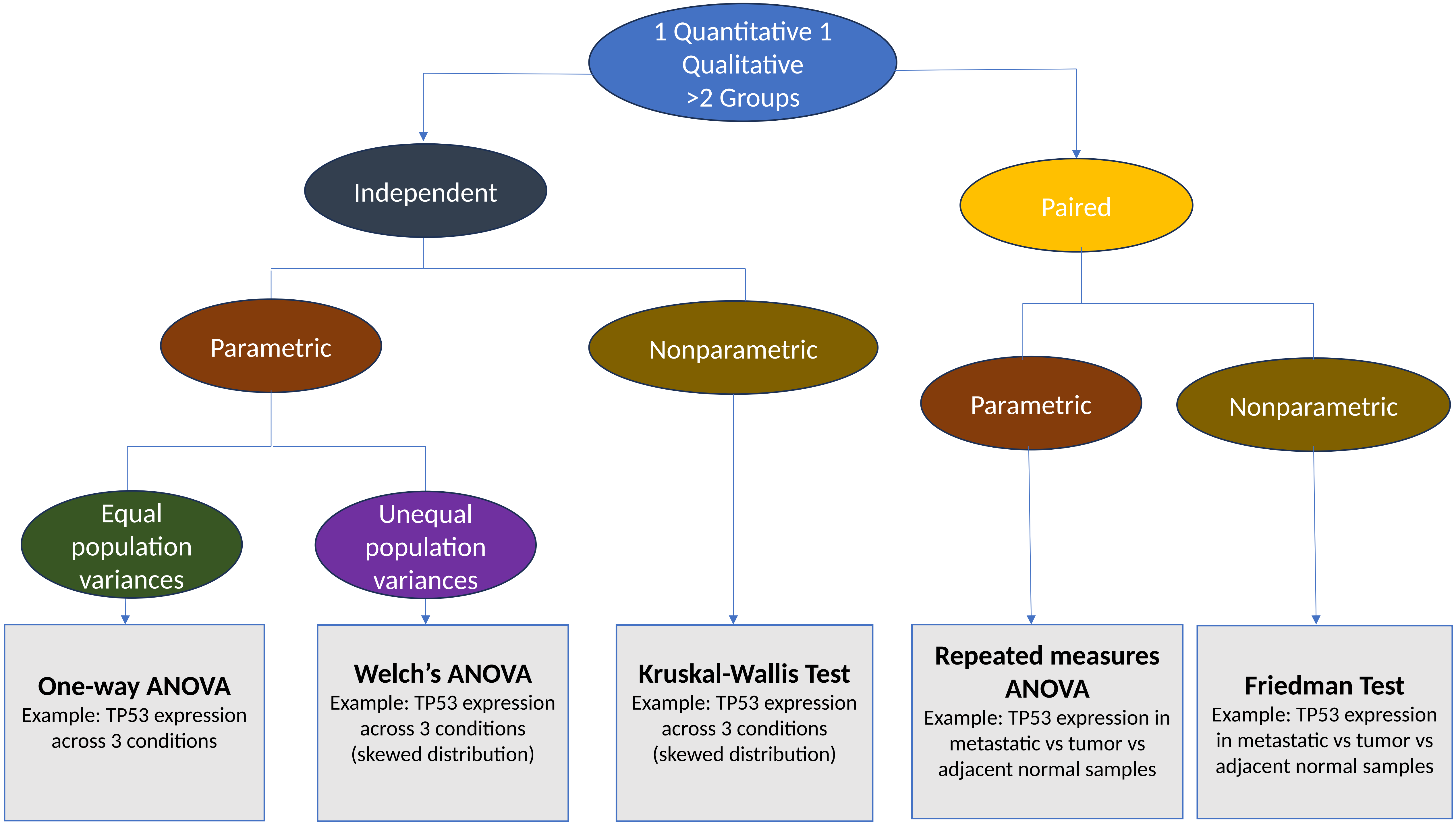

1 Variable
Categorical
Quantitative
2 groups
> 2 groups
Parametric
Non-Parametric
One-Proportion Test
Example: Gene mutation proportion matches expected
Chi-Squared Goodness of Fit
Example: Codon Frequencies against expected frequencies
One-Sample Student's T-test
Example: TP53 expression against known expected expression
One-Sample Wilcoxon Test
Example: TP53 expression against known expected expression (skewed distribution)


>2
Variables
Categorical
Quantitative
2 groups
> 2 groups
Binary Logistic
Regression
Example: Predicting
tumor presence based on
gene expression
Multinomial
Logistic Regression
Example: Predicting
tumor stage based on
gene expression
Multiple Linear
Regression
Example: Prediction
tumor size based on
expression of 4 genes


2
Quantitative
Parametric
Non-
Parametric
Pearson Correlation
Example: Co-expression
with linear relationship
Spearman
Correlation
Example: Co-expression
with non-linear
relationship


2 Qualitative
2 groups
> 2 groups
Paired
Independent
2 Paired Samples
> 2 Paired Samples
Expected frequencies < 5
Expected frequencies < 5
McNemar's Test
Example: Mutation calling differ between 2 different variant callers
Cochran's Q-Test
Example: Mutation calling differ between >2 different variant callers
Fisher's Exact Test
Example: Mutation presence associated with drug response? (n < 5)
Chi-Square Test of Independence
Example: Mutation presence associated with drug response? (n <>5)


1 Quantitative
1 Qualitative
2 Groups
Independent
Paired
Parametric
Nonparametric
Parametric
Nonparametric
Equal population variances
Unequal population variances
Student’s t-test
Example: TP53 expression in tumor vs normal samples
Welch’s t-test
Example: TP53 expression in tumor vs normal samples
Mann-Whitney U Test
Example: TP53 expression in tumor vs normal samples (skewed distribution)
Paired Student’s t-test
Example: TP53 expression in tumor vs adjacent normal samples
Wilcoxon-Signed rank test
Example: TP53 expression in tumor vs adjacent normal samples


1 Quantitative 1 Qualitative >2 Groups
Independent
Paired
Parametric
Nonparametric
Parametric
Nonparametric
Equal population variances
Unequal population variances
One-way ANOVA
Example: TP53 expression across 3 conditions
Welch’s ANOVA
Example: TP53 expression across 3 conditions (skewed distribution)
Kruskal-Wallis Test
Example: TP53 expression across 3 conditions (skewed distribution)
Repeated measures ANOVA
Example: TP53 expression in metastatic vs tumor vs adjacent normal samples
Friedman Test
Example: TP53 expression in metastatic vs tumor vs adjacent normal samples

References

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- *How should I sequence ATAC-seq libraries?* https://dnatech.ucdavis.edu/faqs/how-should-i-sequence-atac-seq-libraries
- *Differential ATAC-seq and ChIP-seq peak detection using ROTS* https://academic.oup.com/nargab/article/3/3/lqab059/6313252
- *Pipeline and Tools for ChIP-Seq Analysis* https://www.cd-genomics.com/pipeline-and-tools-comparison-for-chip-seq-analysis.html
- *Chromatin accessibility profiling by ATAC-seq* https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9189070/
- *Statistics flow chart* https://statsandr.com/blog/what-statistical-test-should-i-do/